

GNSS TEC based ionospheric anomalies related to the 2019 Mw 6.2 Thailand earthquake

by

Shireen Mushtaq

Supervisor

Dr. Munawar Shah

Department of Space Sciences



Institute of Space Technology, Islamabad, Pakistan IST 2022

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A Thesis Submitted to the Institute of Space Technology

in partial fulfillment of the requirements for the degree of Master of Science in Global
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Shireen Mushtaq

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Dr. Munawar Shah

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Shireen Mushtaq

APPROVAL BY BOARD OF EXAMINERS

Dr. Munawar Shah

Internal Examiner

External Examiner

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ABSTRACT

Detecting seismic ionospheric perturbation by Global Navigation Satellite System (GNSS) is an inevitable approach and has proven to be a new field in the investigation of Earthquake (EQ) precursors. In this paper, the temporal Total Electron Content (TEC) perturbation are examined from the International GNSS Service (IGS) stations round the epicenter of the November 20, 2019, Thailand EQ (M_w 6.2), that occurred on UTC=23:50:43 h. Seismo ionospheric perturbations are estimated with TEC during thirty-one days (twenty days prior and ten days followed by EQ). The solar and geomagnetic indices are checked in the context of EQ. Quiet geomagnetic activity is observed on the day of EQ. However, K_p is larger than 3 immediately after the EQ. Results depict the TEC perturbation measured at three GNSS stations. Furthermore, clear positive anomalies are detected within three days prior to EQ M_w 6.2, (-3, -2, and -1 days) during quiet geomagnetic activities. Since, $K_p > 3$ after the EQ. Therefore, the ionospheric perturbation is related to geomagnetic storms.

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LIST OF ABBREVIATIONS

GNSS	Global Navigation Satellite Systems
GPS	Global Positioning System
GLONASS	Global'naya Navigatsionnaya Sputnikova Sistema
BDS	BeiDou Navigation Satellite System
USA	United States of America
PNT	positioning , navigation and timing.
TEC	Total Electron Content
Ap	Planetary A-index
Kp	Planetary K-index
AE	Auroral Electrojets
Dst	Disturbance storm-time
EQ	Earthquake
IGS	International GNSS Services
USGS	United States Geological Survey
ISGI	International Services for Geomagnetic Indices
STEC	Slant TEC
VTEC	Vertical TEC
UB	Upper Bound
LB	Lower Bound
UT	Universal Time

1 INTRODUCTION TO GNSS

Global Navigation Satellite System (GNSS) is the general word for all of the Satellite Systems that is working or under construction. It can be categorized as a vastly accurate, continuous, all weather and L-band method with signals over the Earth's atmosphere. GNSS is a system that can deliver positioning and timing services through ranging from a receiver to the satellites.[1] Generally, a GNSS has 3 segments that is the space ,ground and the user segment. The space segment comprises of constellation of satellites. The satellites transmit the signals that comprise the range measurements data. The ground segment contains the worldwide spread monitoring stations to observe the working status of satellites. The user segment is collected of the receivers. The end users getting the signals from the satellites to acquire the locations and time. Individually GNSS satellite constantly transmit the radio signals in two or more than two frequencies in L band with wavelength about twenty cm. These signals are used for PNT.

Now, there are four GNSS systems:

- GPS
- GLONASS
- Galileo
- BeiDou.

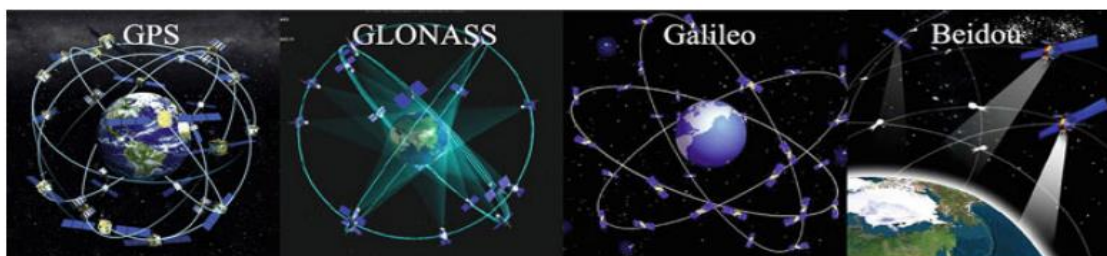


Fig. 1. 1 Global Navigation Satellite Systems

1.1 GNSS History

1.1.1 GPS

In 1960s, many organizations of U.S. (like DOD, NASA, DOT etc.) were concerned in emerging satellite systems for location determination. Transit is first navigation system that constellation consist of 5 satellites[2]. The transit system was used by the US Navy. In 1973, NAVSTAR system was created., which was called GPS later. The GPS is a satellite navigation system worked by the United State. In that time, it just used for military purpose. In 1983, it has been accessible for civilian user. It was become fully operating with 24 satellites in 1995[3]. Since 1973, a number of blocks have been launched. GPS provides precise information on PNT everywhere in whole world with in all type of weather conditions. GPS offers two positioning services(The Precise Positioning Service (PPS) and the Standard Positioning Service (SPS)) [4]. This system contains of 3 segments that is discussing below.

1.1.1.1 The Space Segment:

The SS contains GPS satellites. The spaceship drive radio signals from space. Currently thirty-one GPS satellites rotate around the Earth in MEO. The satellites circulated in six orbital planes with a height of approximately 20,200 km from the Earth surface, with 55-degree inclination to the equator and orbital period about 11 hours 58 minutes. Total sixty-seven GPS satellites successfully launched into orbit from 1978. [5][6] GPS satellites always broadcast their identity, ranging alerts, satellite fame and corrected ephemerides. The satellites are recognized by their Pseudo Random Code Number.

1.1.1.2 The Control Segment:

The CS contains tracking stations situated everywhere in the world. The CS consist of master station, substitute master station control, 4 ground antennas and 6 monitor stations.

[6]

1.1.1.3 The User Segment:

The US contains receivers and user civic. The receivers convert satellite signals into PVT estimations.

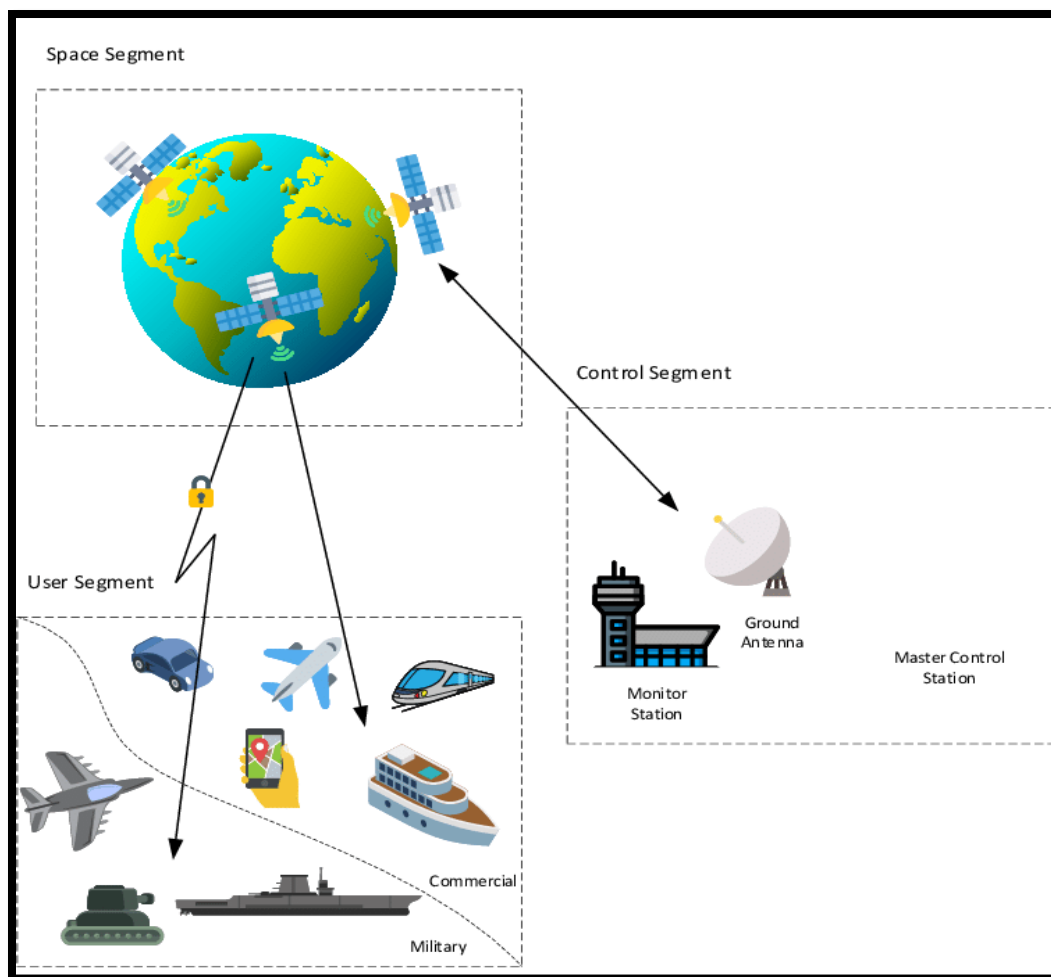


Fig. 1. 2 GPS Segments

1.1.1.4 GPS Signals:

The GPS satellite broadcast data on 2 frequencies, L1 and L2 and the range are 1.57542 and 1.22760 in GHZ respectively. The satellite's atomic clocks generate the L-band frequency. The L1 and L2 carrier frequencies are produced by multiply the L-band frequency (10.23 MHz) by 154 and 120, respectively. The GPS signal is composed of three parts: Carrier, Navigation data and Satellite codes. The GPS system broadcast two signals, a C/A code, precise encrypted code that is existing for civilian uses and for military uses respectively. All satellites of GPS broadcast by using the CDMA technique. [7][6]

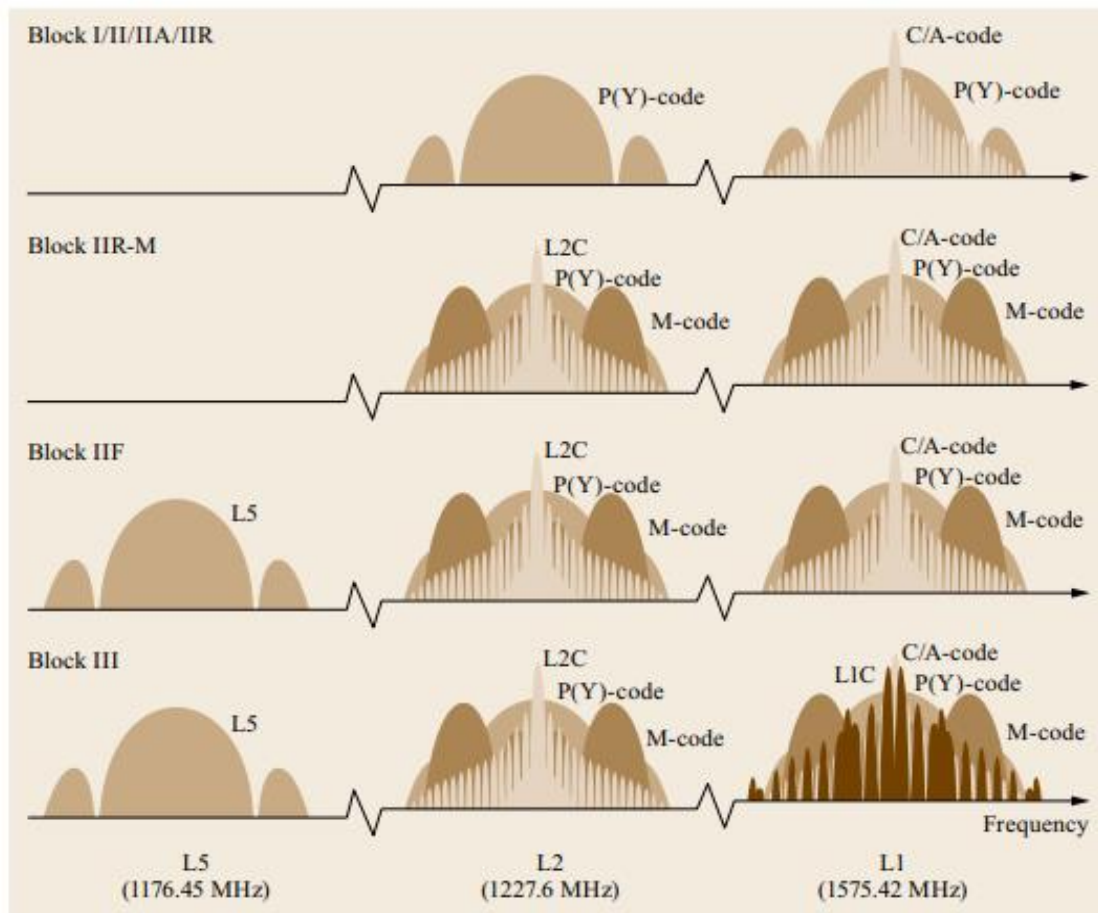


Fig. 1. 3 Evolution of the GPS signals

Table 1. 1 GPS

GPS		Country	United States of America
Official Website Address		https://www.gps.gov/	
No. of Orbits	6	Reference Time System	UTC
Inclination of Orbits	Inclination of 55° in relation to the equator	Terrestrial Reference System	World Geodetic System WGS-84
No. of Satellites in Each Orbit	Four satellites	Coverage Area	Global
Types of Orbits	MEO	Positioning Accuracy	500–30 cm
Total Satellites in constellation	30 operational GPS satellites	Timing Accuracy	≤40 nanoseconds
Orbital Height	20,200 km	Signal Types	L1 , L2 , L5
Location of Master Station	Located at Colorado Springs	Signal Frequencies	• L1(1.57542 GHz) • L2 (1.2276 GHz) • L5 (1.17645 GHz)
No. of Ground Stations	16	Modulation	BPSK

1.1.2 GLONASS

This system was established through the Russian Federation. This program started in 1976 and the 1st satellite was launched in 1982. It was stated completely operating in 1995. Unluckily, the full constellation was short existed. In 2002 the constellation had fell. With the help of Russian government, this system was renewed, and in December 2011, It was stated fully operating. GLONASS satellite system offers the positioning and timing data to users.[8] GLONASS gives dual frequency L band navigation signals that's for civil and military navigation. Currently GLONASS satellite system consist of 24 satellites that is revolve around the Earth. The satellites circulated in three orbital planes with height of approximately 19,130 km from the Earth surface, with 120-degree inclination to the equator and the orbital period of 11 hours 15 minutes 44 seconds equivalent to 8/17 of a sidereal day. The GLONASS system broadcasts the signals in two bands: L1 (1.602 GHz), L2 (1.246 GHz) and L3 (1.202GHz). [9]

The GLONASS constellation give visibility to number of satellites.

A least 4 satellites permit a GLONASS receiver to compute its location in 3D and to match with system time.

The GLONASS system are characterized in 3 generations:

- 1st generation GLONASS I/II
- 2nd generation GLONASS-M
- 3rd generation GLONASS-K

The GLONASS system contains of three segments:

1.1.2.1 The Space Segment(SS):

The SS consist of GLONASS satellites that give the information about GLONASS ranging.

The signal of GLONASS satellites classifies the satellites and contains:

- information about satellite health
- Almanac of all other GLONASS satellites.
- PVT information for computing satellites locations.[10]

1.1.2.2 The Control Segment:

The CS includes of the system control center and a network of command tracking stations throughout Russia. The CS monitor the status of satellites and upload the navigation data to the satellites two time in a day.[10]

1.1.2.3 The User Segment:

This segment comprises of equipment that tracks and gets the signal of the satellites.

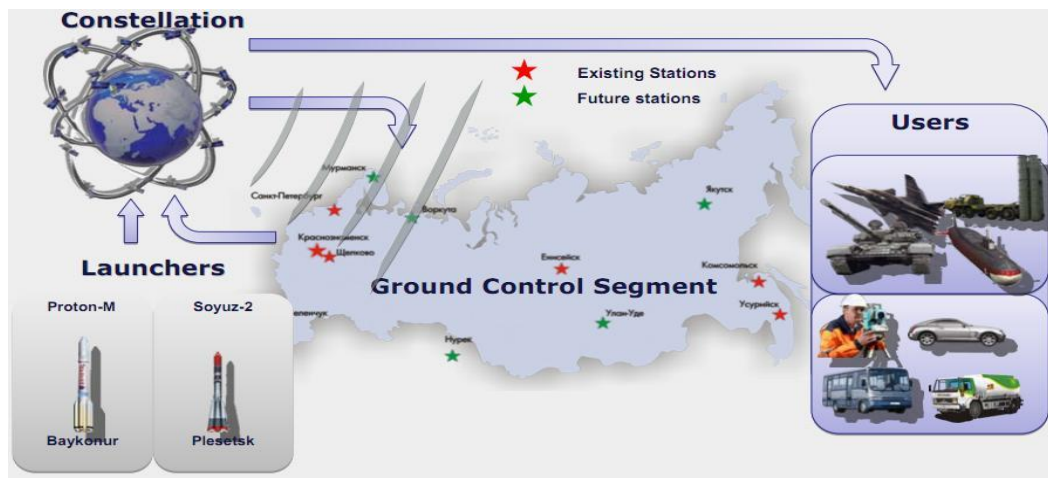


Fig. 1. 4 GLONASS segments

Table 1. 2 GLONASS

GLONASS		Country	Russia
Official Website Address		https://www.glonass-iac.ru/en/	
No. of Orbits	3	Reference Time System	UTC
Inclination of Orbits	64.8 degree	Terrestrial Reference System	PZ-90.11
No. of Satellites in Each Orbit	8	Coverage Area	Global
Types of Orbits	MEO	Positioning Accuracy	5–10 m
Total No. of Satellites in constellation	28	Timing Accuracy	200 ns
Orbital Height	19,140km	Signal Types	L1, L2, L3
Location of Master Station	Krasnoznamensk	Signal Frequencies	L1 1.602 GHz (SP) L2 1.246 GHz (SP) L3 1.202GHz
No. of Ground Stations	22	Modulation	BPSK

1.1.3 GALILEO

This system is recognized by the European Union. The 1st and 2nd IOV satellites were launched on October ,21 2011, and the 3rd and 4th IOV satellites had been released on October,12 2012. Currently GALILEO satellite system comprise of 27 satellites that is revolve around the Earth. The satellites circulated in three orbital planes with height of about 23,222 km from the Earth surface, with 56-degree inclination to the equator and the orbital period of 14:04:45. The constellation give the global coverage with 6 satellites in sight from the Earth at any time of day. [11]

The GALILEO system consists of three segments:

1.1.3.1 Space Segment:

This segment comprises of thirty satellites distributed equally and regularly in 3 orbit planes. The estimated height is slightly greater than from GPS 23,222 km with 56-degree inclination to the equator.

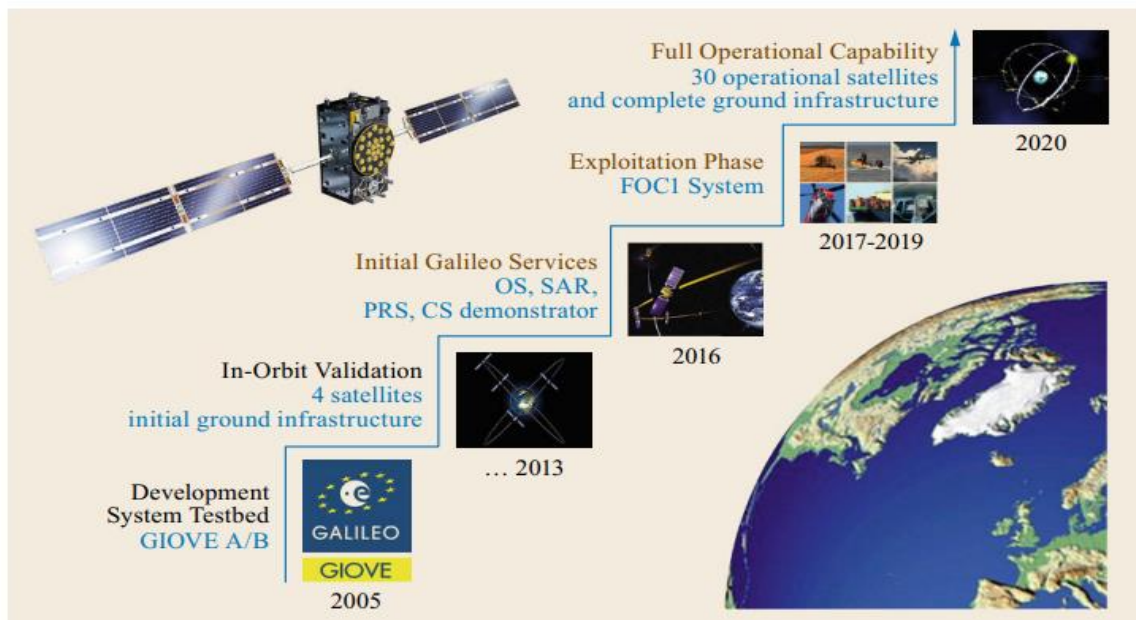


Fig. 1. 5 GALILEO deployment (courtesy of M. Pedoussaut, S. Corvaja, Th. Burger, ESA, and pixabay.com)

1.1.3.2 Control Segment:

The CS is accountable for:

- working the constellation of navigation satellites.
- monitoring main purposes of the navigation operation.
- clock synchronization.

The CS consist of:

- twelve to fifteen reference stations
- five up link stations
- two control centers.

1.1.3.3 User Segment:

This segment include user receivers, with different capabilities associated to signals for accomplish the services.[11]

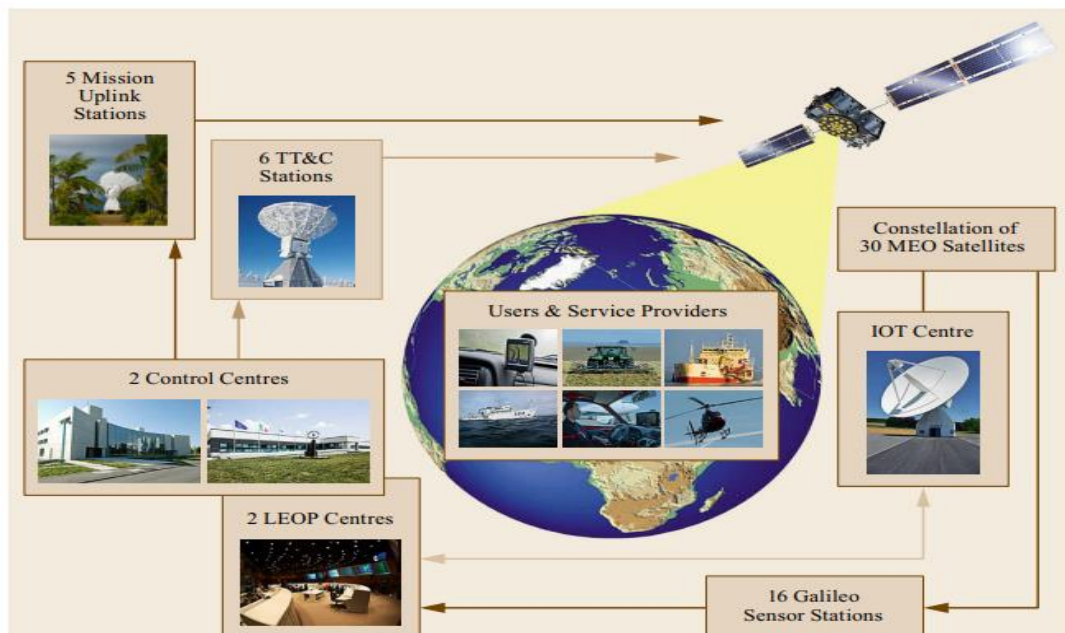


Fig. 1.6 Galileo system overview (courtesy of ESA/M.)

Table 1. 3 GALILEO

GALILEO		Country	European Union			
Official Website Address		https://ec.europa.eu/growth/sectors/space/galileo_en				
No. of Orbits	3	Reference Time System	Galileo System Time (GST)			
Inclination of Orbits	56° to the equator	Terrestrial Reference System	Galileo Terrestrial Reference Frame (GTRF)			
No. of Satellites in Each Orbit	10 satellites in each orbit	Coverage Area	Global			
Types of Orbits	Medium Earth Orbit	Positioning Accuracy	1 meter (public) 1 cm (encrypted)			
Total No. of Satellites in constellation	30	Timing Accuracy	30 ns			
Orbital Height	23,222 kilometers	Signal Types	E1, E5, E6			
Location of Master Station	Prague, Czech Republic	Signal Frequencies And Modulation	Parameters	Modulation	Frequency(MHz)	
			E1-I	CBOC (6,1,1/11)	1575.42	
			E1-Q	BOC (15,2.5)	1575.42	
			E5a	AltBOC (15,10)	1176.45	
			E5b	AltBOC (15,10)	1207.14	
			E6-I	BPSK (5)	1278.75	
			E6-Q	BOC(10,5)	1278.75	

1.1.4 BEIDOU:

The BEIDOU System has been individually built and worked by China. This system gives all time, all weather and high accuracy PNT services to worldwide users. BEIDOU has twenty-two Operating satellites in orbit. BeiDou has 2 distinct constellations. BeiDou-1 is the 1st generation that was consist of three satellites. It developed working in 2000 and accessible restricted analysis and navigation services, generally for users that are living in China and around the regions. Beidou-1 was out-of-service toward the finish of 2012. BeiDou-2 is the second era of the framework. It became employable in the 2011 with a partial constellation of ten satellites within the circle. Up and coming age of it's far BeiDou-3. The principal BDS-3 satellite was sent off in March 2015. As of January 2018, nine BDS-3 satellites have been sent off. BeiDou-3 was relied upon to be completely useful before end of 2020.[12][13] This system is consisting of three segments:

1.1.4.1 A Space Segment:

This segment of BDS comprises of a number of satellites positioned in the GEO, IGSO and MEO.

1.1.4.2 A Ground Segment:

This segment of BDS comprises of many ground stations, with master control stations, time synchronization/uplink stations and monitoring stations. [14]

1.1.4.3 A User Segment:

This segment of BDS comprises of many types of BDS basic products, structures, and offerings together with different navigation systems.[13]

Table 1. 4 BeiDou

BeiDou				Country	China		
Official Website Address				http://en.beidou.gov.cn			
No. of Orbits	GEO	1		Reference Time System	UTC		
	MEO	3					
	IGSO	3					
Inclination of Orbits	55°			Terrestrial Reference System	CGCS 2000		
No. of Satellites in Each Orbit	Three IGSO orbits, with one IGSO per plane. Twenty four MEOs in three planes plus three spares.3 GEO			Coverage Area	Global		
Types of Orbits	GEO,MEO,IGSO			Positioning Accuracy	10 meters		
Total No. of Satellites in constellation	35 satellites,			Timing Accuracy	50 ns		
Orbital Height	GEO	35 786 km		Signal Types	B1, B2, B3		
	MEO	21 528 km					
	IGSO	35 786 km					
Location of Master Station	Beijing			Signal Frequencies	B1	1.561098 GHz	
					B1C	1.57542 GHz	
					B2a	1.17645 GHz	
					B3	1.268520 GHz	

1.2 GNSS Applications:

The GNSS receivers, are used in various applications. The first systems were established in twenty century, mostly to relief military employees to discover their way, surveying and mapping.[15] But today, GNSS is use in several applications for military and civilian use. It is used in agriculture, transportation, machine control, marine navigation and other manufacturing where efficiencies can be increased from the application of accurate, frequently existing location and time data. The GNSS is furthermore used in a comprehensive variety of consumer applications, with vehicle navigation, mobile communications, hiking etc. Some application is following.

1.2.1 Consumer:

GNSS application has been accepted through consumer market, in a wide-ranging product. The receivers of GNSS are nowadays regularly combined into mobile communications equipment, to support application that show maps viewing the position of and finest way to stores and restaurants. [16][17]

1.2.2 Transportation:

There are many applications in this field like rail transportation, aviation, marine transportation and surface transportation. In the rail transportation, the GNSS is used in combination with other technologies, to track the position of engines and rail cars. In this way we can check the accurate position of rail, delays and operational expenses, improving protection, track capability and consumer facilities. In aviation, the GNSS is used for airplanes navigation (like departure, enroute and landing). In marine transportation the GNSS is used for precisely determine the location of craft when the crafts are in the sea.

The GNSS is integrated into subaquatic measuring, navigation danger position and mapping.[18][19]

1.2.3 Machine Control:

The machinery like graders, pavers, bulldozers, excavators etc. are work on the basis of GNSS technology. There are three major areas of this application are surface mining, construction and agriculture. The benefits of this application are efficiency, precision, data management and theft detection.[20]

1.2.4 Geospatial Information system (GIS)

A GIS captures, stores, examines manages and current information that is related to position. This application is used in many companies (like insurance, utility, municipal planning companies etc.). It can be creating different types of maps from the data and convert in to digital map.[21]



Fig. 1. 7 Application of GNSS

1.3 Ionosphere and GNSS

In GNSS system, the transmission speed of a navigation signal is most essential. we can calculate the ranging by multiply the time and speed when we know the signal's propagation time as of the satellite to the receiver.[22] Electromagnetic signal's propagation speed is nearly equivalent to $c = 3 \times 10^8$ in vacuum case. The GNSS systems transmit the radio signals that can be get significant effects in their transmission concluded the Earth's atmosphere to the receiver.[23] The free electrons, molecules ions that exists in the ionosphere's layers that affects the propagation speed and direction of signals.

1.3.1 Layers of ionization:

The ionospheric areas are overlying height dependent layers. The ionosphere lies between the height of about 50 to 1,000 km The layer of ionosphere comprises of a height of maximum density. The ionization's layer is further consist of D, E and F layers. [23]

1.3.1.1 The D Layer

This layer is lowermost layer of the ionosphere and it is around fifty to ninety km in altitude from surface of the Earth. The lesser part of this region is stated as the C layer where ionization formed through cosmic rays, and it causes abrupt increase of electron density in sunrise and quick decrease at sunset. The frequency of collision among electrons and molecules in this layer is high in the daylight.[22] The free electrons in this layer is less than the number of molecules. The absorption is less at nighttime and maximum around noontime. The D layer vanish at nighttime afterward sunset.

1.3.1.2 The E layer

This layer is the next layer afterward the D layer and it began from approximately ninety to one forty km. The behavior of this layer totally depends on solar activity and the zenith angle of the sun. This layer is free of conflicts unlike other layers. The sun's X-ray is the main source of ionization that is emissions resultant in electron densities presenting different solar-cycle, seasonal and daily variations. In the night this layer starts to weaken steadily as the main source of ionization is the Sun that is no longer there, but the layer does not completely vanish. This layer does not entirely disappear at night. The electron densities for mid latitude noon time are among 1.3×10^{11} and 1.7×10^{11} electrons/m³. The dense layer (Es) layers do arise with different mechanism sources at middle and lower latitudes also the polar area.[24]

1.3.1.3 The F layer

The F layer began from approximately one forty to one thousand km altitude. The F layer is the uppermost layer of the ionosphere. This layer comprises of just one layer in the night, but during the daytime it is divided into two layers. The F1 layer is the lowest part of F layer during the daytime, and it is from approximately 140 to 300 km from the Earth surface. It occurs only in day-time and vanish at night. Most of density for the F1 layer happens just after noon. The F2 layer is situated nearby the peak electron density and it is from approximately 300 to 400 km. This layer has maximized rate of ion production due to less number of neutral particles. The main cause of ionization in F layer is the photo ionization of atomic oxygen. In daytime the maximum concentration of electrons can be seen and the concentration decreases at nighttime.[25]

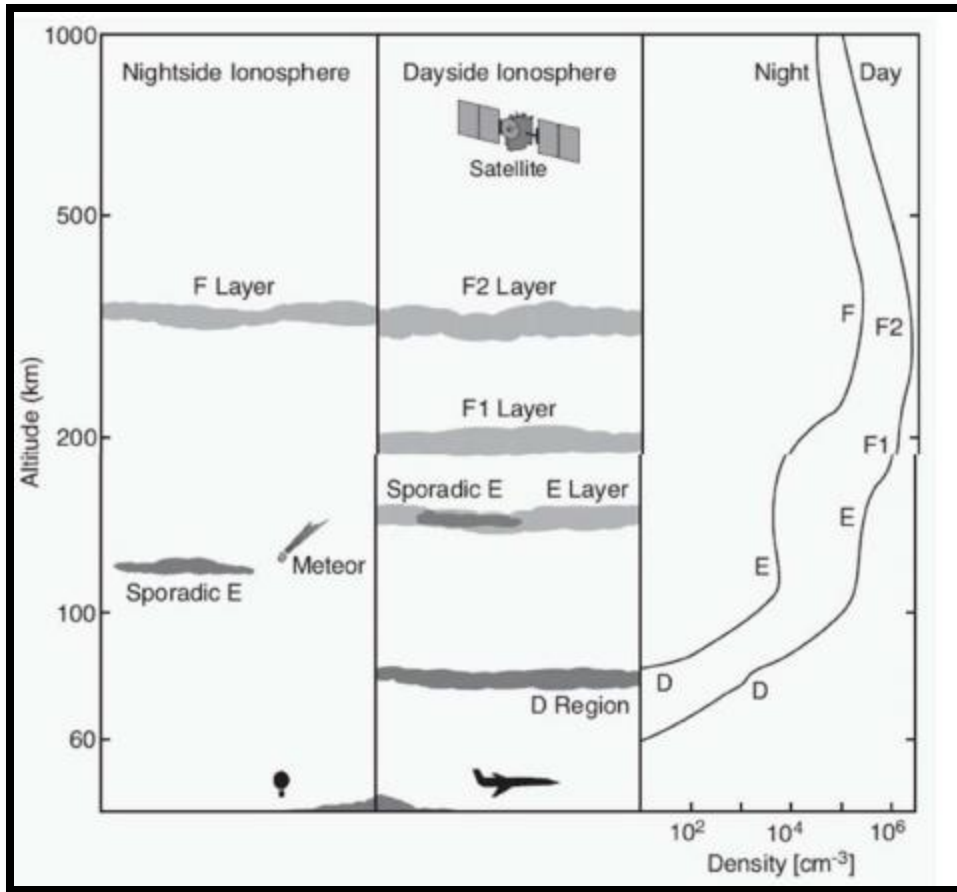


Fig. 1. 8 ionosphere layers with electron density

1.3.2 Ionospheric Model

The ionosphere is the portion of the atmosphere that start from approximately fifty km and end at approximately 1000 km. It comprises a moderately ionized medium, occurrence of charged particles and x rays and UV rays. Free ions that is present in this layer to disturb the radio waves. Group delay and phase advance occur while the signal go through the ionosphere layers. Now, the Ionospheric models like the Klobuchar model used in GPS and BeiDou and the NeQuick G model used in Galileo. The Ionospheric model represents an explanation of an ionospheric parameter (the electron density, in 3D and time). [26]

1.3.2.1 Klobuchar Ionospheric model

This Ionospheric model was considered by Klobuchar in 1987 for single frequency GPS users. [27] It is estimated to decrease approximately the fifty percent RMS ionospheric range error overall. In Klobuchar model, the ionospheric time delay is continuous in the nighttime, that's equivalent to five ns, as well as cosine function happening the day. In this model, the daytime variant of the ionospheric interruption is just by mean of a half cosine feature by changing in amplitude and length, this is relying on time and range of geomagnetic. (Fig. 1.8) To calculate the amplitude and period in any Earth position, 8 coefficients of 2/3 degree polynomials, are broadcasted every day in the navigation messages of the GPS. The GCS of the GPS computes and informs about constants rendering to season and lunar action level.[28]

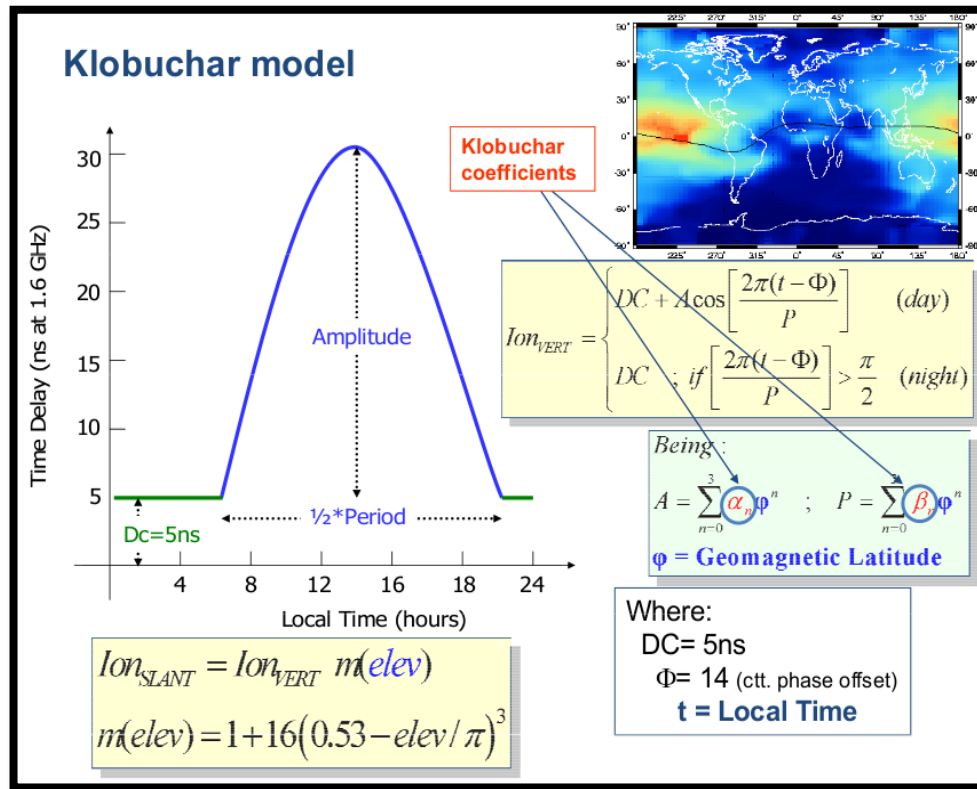


Fig. 1. 9 Klobuchar ionospheric model. Algorithm layout

1.3.2.2 NeQuick Ionospheric Model

This Ionospheric Model implemented in the Galileo system. This model is used for single-frequency users to calculate the ionospheric delay corrections. It is a 3D and time dependent partially empirical electron density model. There are 3 core ionospheric layers like E, F1, and F2 are use in this. Complex algorithm is used in NeQuick Ionospheric Model than Klobuchar. [29] [30] Galileo satellites broadcast navigation message contain 3 ionospheric coefficients that are used to calculate the A_z as follows

$$A_z = a_0 + a_1u + a_2u^2 \quad (i)$$

where the coefficients a_0, a_1, a_2 are enhanced each twenty-four hours the usage of a fixed of worldwide permanent stations and broadcasted to the users. And μ is the modified dip latitude.

$$\tan u = \frac{I}{\sqrt{\cos \emptyset}} \quad (ii)$$

Where I is the real magnetic inclination and $\emptyset$ the terrestrial latitude of the receiver.[31]

1.4 Total Electron Content

TEC is describing as the total number of electrons along a path among the satellite and Earth station. TEC is basic parameter, which conclude the main things of the Ionosphere. The number of free electrons along the ray path of $1m^2$ cross section that extends all the way up from the ground through the ionosphere is referred as TEC.[32] It is computed in expressions of TEC unit, and 1 TECU is equal to $10^{16} e/m^2$. The range of TEC is about 10^{16} to 10^{19} with least and highest happening at middle of the night and midafternoon. At nighttime the TEC decays quite slowly because the recombination of electrons and ions.

Maximum TEC generally happens in the early afternoon and smallest TEC generally happens just earlier daylight. As well day-to-day TEC variants rise, when the sun is directly. The measurement of TEC is significant for the improvement of propagation effects that is applied on radio systems. The measurement of TEC is use for EQ observing, modelling and prediction. TEC is the line integral of electron density on desired ray path. [33] TEC can be measured with different ways (ionosondes, incoherent backscatter radars, GNSS etc.). Excepting the GNSS signals, the other ones are very costly. The extensive spread use of GPS dual frequency receivers delivers a cost effective result to estimate the TEC. There are two types of TEC: VTEC and STEC. STEC is monitored by the GPS satellite. The STEC is evaluated via following equations

$$STEC = \frac{f_1^2 f_2^2}{40.28(f_2^2 - f_1^2)}(l_1 - l_2 + \lambda_1(N_1 + b_1) - \lambda_2(N_2 + b_2) + \epsilon) \quad (\text{iii})$$

$$STEC = \frac{f_1^2 f_2^2}{40.28(f_2^2 - f_1^2)}(P_1 - P_2 - (d_1 - d_2) + \epsilon) \quad (\text{iv})$$

where STEC is Slant TEC, f_1 and f_2 denotes carrier wave frequencies, λ_1 and λ_2 denotes wavelengths, l_1 and l_2 are the carrier phases, N_1 and N_2 depicts the uncertainties of the signal, b_1 and b_2 symbolizes the carrier phases of the signal, P_1 and P_2 denotes the pseudo-ranges of the signal, and ϵ is the random residuals of the signal.[34] To transform STEC into VTEC, the below equation is used.

$$VTEC = STEC \times \cos (\arcsine (R \sin Z / (R + H))), \quad (\text{v})$$

where the Earth radius is represented by R , H is the IPP ionospheric height, and Z represents the zenith angle of the satellite.

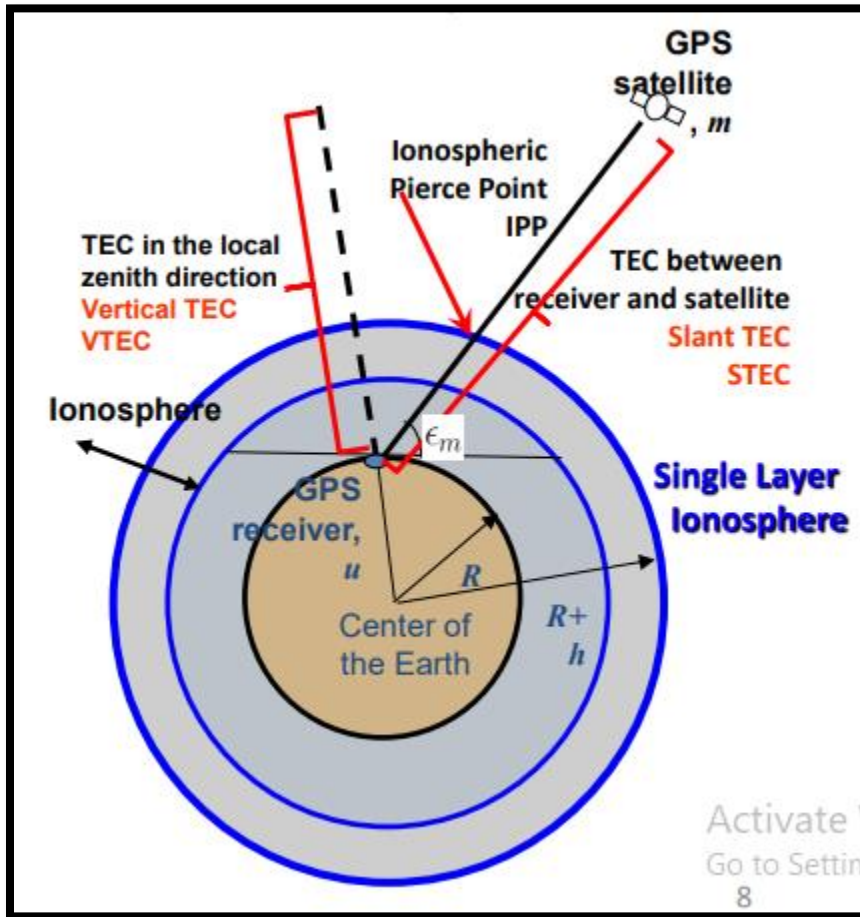


Fig. 1. 10 Total Electron Content Measurement.

1.5 Motivation and Objectives

Since the signals of GNSS give a profound understanding above ionospheric conditions by assessing the measure of TEC. In this thesis we study the ionosphere and its performance in the seismic activity. GNSS can assist us with observing ionosphere and to examine ionospheric activity which might help in considerate the EQ-ionosphere associations. Along these lines, it turns out to be extremely challenging to connect an ionospheric variation with the EQ at some stage in solar or geomagnetic activity. In this thesis, I will examine GNSS TEC based ionospheric anomalies related to the 2019 Mw 6.2 Thailand

earthquake. The TEC values are acquired from the GNSS stations functioning around the EQ. The selection conditions will be:

- i. There must be three GNSS stations functioning nearby the EQ breeding zone.
- ii. The EQ must happen in the quiet geomagnetic activity.

All the investigation, methods, and figures will be accomplished in the MATLAB® 2016.

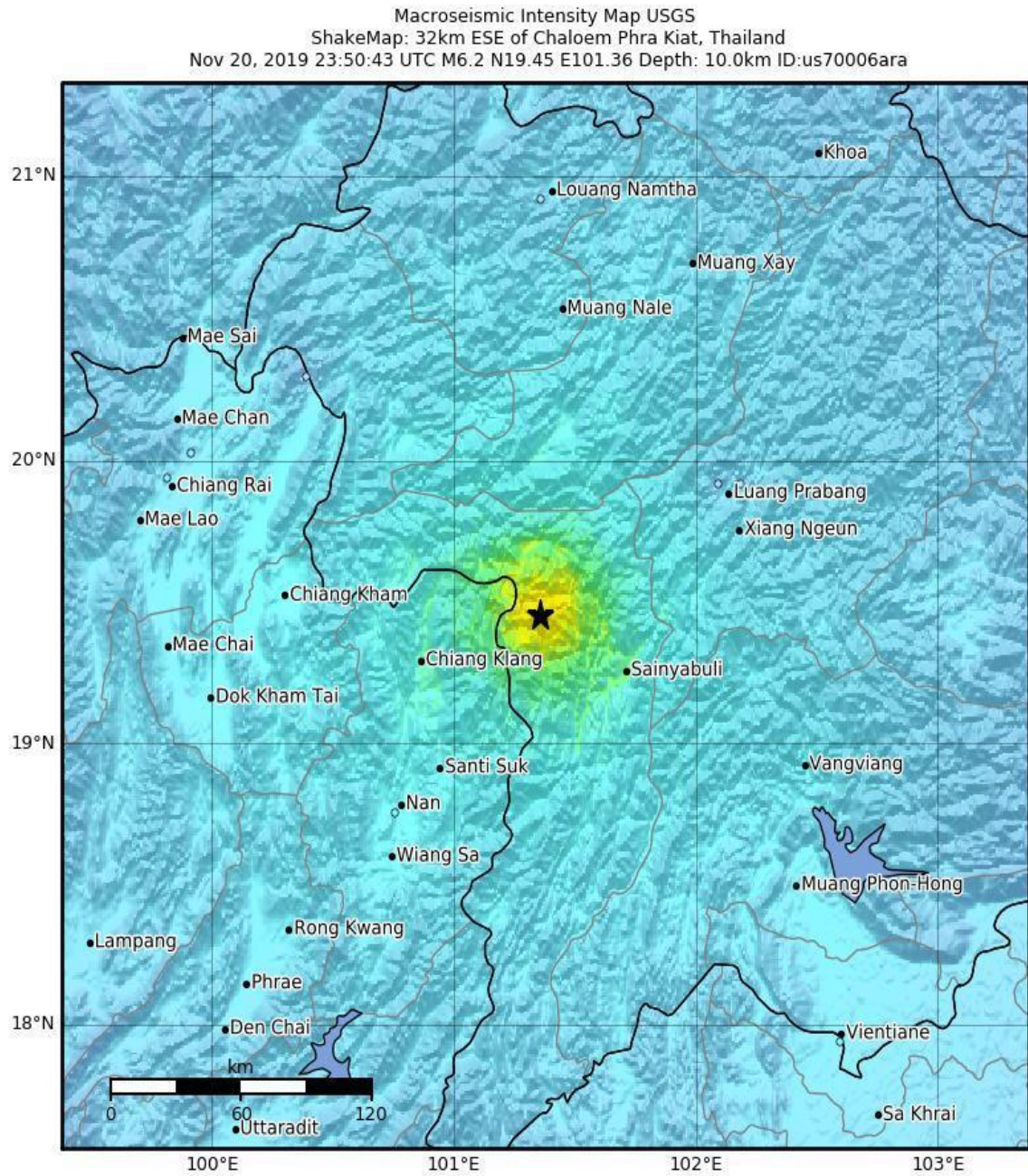
2 DATA AND METHODOLOGY

2.1 Case Study

In my research work, the ionospheric conditions are studied afore the EQ of November 20, 2019 in Thailand (M_w 6.2) happened on UTC = 23:50:43h at the geographical location of Latitude (19.45°N) and Longitude (101.36°E), and shallow hypo central depth 10 km according to United States Geological Survey (USGS). According to Global Disaster Alert and Coordination System or GDACS (<https://gdacs.org/report.aspx?eventid=1194401&episodeid=1279805&eventtype=EQ>), there are no reports of casualties or damage. Some people near the epicenter and surrounding areas felt aftershocks throughout the morning period. Residents of Chiang Mai, Thailand have long felt the shakes but no major damage. Up to 8,000 people were exposed to strong shaking. To examining the anomalies in the ionosphere for selected Earthquake, it is mandatory to distinguish seismic ionospheric perturbation with solar activity between the EQ era. For this purpose, geomagnetic indices are being analyze in this thesis. So, to check the possible EQ induced ionospheric anomalies in November 2019, we distinct selection criteria that is based on these conditions. The 1st condition is: the chosen EQ have to show up in a quiet geomagnetic activity period ($K_p \leq 3$, $Dst \geq -20$). The 2nd condition is: there have to as a minimum 3 IGS stations falling across the EQ preparation zone.

Table 2. 1 The information for the selected EQ in Thailand.

No.	Date	Time(UTC)	Latitude($^\circ\text{N}$)	Longitude($^\circ\text{E}$)	M_w	Depth(km)
1.	20/11/2019	23:50:43	19.45	101.36	6.2	10



SHAKING	Not felt	Weak	Light	Moderate	Strong	Very strong	Severe	Violent	Extreme
DAMAGE	None	None	None	Very light	Light	Moderate	Moderate/heavy	Heavy	Very heavy
PGA(%g)	<0.0464	0.297	2.76	6.2	11.5	21.5	40.1	74.7	>139
PGV(cm/s)	<0.0215	0.135	1.41	4.65	9.64	20	41.4	85.8	>178
INTENSITY	I	II-III	IV	V	VI	VII	VIII	IX	X+

Scale based on Worden et al. (2012)

Version 1: Processed 2020-06-06T02:57:19Z

△ Seismic Instrument ○ Reported Intensity

★ Epicenter

Fig. 2. 1 Shows Shake map of seismic event. [Courtesy: USGS]

2.2 Method:

2.2.1 Geomagnetic Indices Data

Different geomagnetic indices (i.e., Dst, Kp, Ap and F10.7) are analyzed to verify the space weather conditions (SWC) and quiet days before and after the EQ. For this task, the source data during October 31-November 30, 2019 are obtained from International Services of Geomagnetic Indices (ISGI) (http://isgi.unistra.fr/data_download.php) and OMNI web NASA <https://omniweb.gsfc.nasa.gov/form/dx1.html>.

2.2.2 GNSS-TEC measurement

In this thesis, we examine the TEC anomalies from GNSS TEC record. The perturbation occurs in ionosphere due to earthquake energy that release from the earth. The TEC is a main source for determining the ionospheric anomalies. [35] The GNSS TEC can be examined in two terms:

➤ Slant TEC

➤ Vertical TEC

STEC is monitored by the GPS satellite. The STEC is evaluated via following equations

$$STEC = \frac{f_1^2 f_2^2}{40.28(f_2^2 - f_1^2)}(l_1 - l_2 + \lambda_1(N_1 + b_1) - \lambda_2(N_2 + b_2) + \epsilon) \quad (\text{vi})$$

$$STEC = \frac{f_1^2 f_2^2}{40.28(f_2^2 - f_1^2)}(P_1 - P_2 - (d_1 - d_2) + \epsilon) \quad (\text{vii})$$

where

- STEC is Slant TEC
- f_1 and f_2 denotes carrier wave frequencies
- λ_1 and λ_2 denotes wavelengths
- l_1 and l_2 are the carrier phases

- N_1 and N_2 depicts the uncertainties of the signal
- b_1 and b_2 symbolizes the carrier phases of the signal
- P_1 and P_2 denotes the pseudo-ranges of the signal,
- ϵ is the random residuals of the signal.

To transform STEC into VTEC, the below equation is used.

$$\text{VTEC} = \text{STEC} \times \cos (\arcsine (R\sin Z/(R+H))) \quad (\text{viii})$$

where

- R is the radius of the Earth
- H is the IPP ionospheric height
- Z represents the zenith angle of the satellite.[36]

The daily GNSS TEC data are retrieved from the IGS stations within EQ geographical position. Furthermore, the TEC data are obtained via this website <http://www.ionolab.org/webtec/single.html?locale=en>. These IGS stations are in operation during the resultant EQ preparation zones. The EQ radius is calculated by Dobrovolsky equation as shown in Eq. (1).[37]

$$\dot{R} = 10^{0.43 \cdot M} \quad (\text{ix})$$

where

The radius of seismic region is represented by $\dot{R}$ (unit: km)

M is the EQ magnitude.

Here, the M is 6.2, hence, the EQ radius is about 463.5 km as depicted in Fig. 2.2

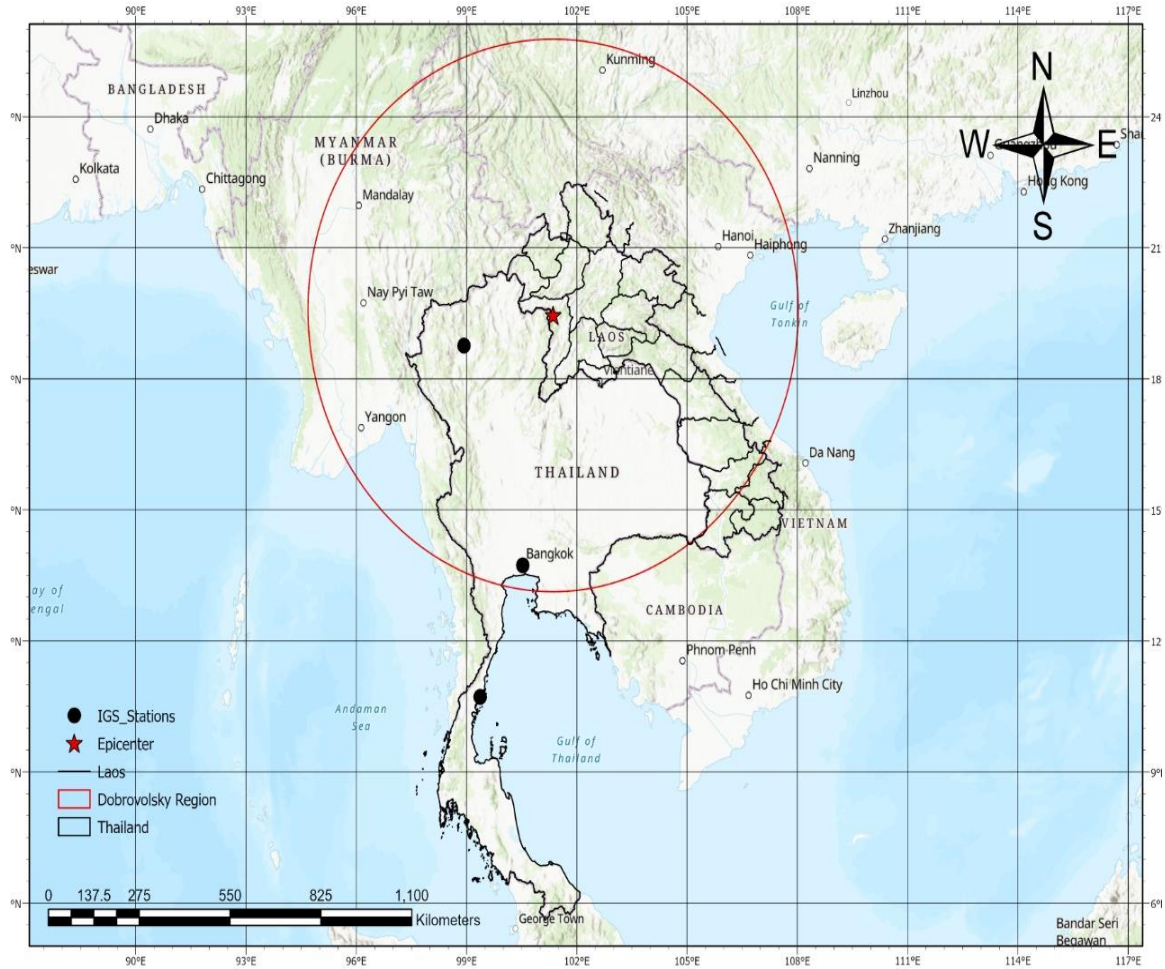


Fig. 2. 2 The EQ Location (Mw 6.2) is denoted by red filled star. The black color filled circles denote the GNSS stations, while the red circle is the preparation zone estimated by Dobrovolsky equation, which is about 463.5 km.

Table 2. 2 GNSS TEC station data of the 2019 Mw 6.2 Thailand EQ

S. No.	GNSS TEC Stations	Geographic	Coordinates
		Latitude (°)	Longitude (°)
1.	CMUM	18.761	98.932
2.	CUUT	13.736	100.534
3.	CPNM	10.725	99.374

2.2.3 Positive and Negative Anomaly

TEC values that lie above the upper bound show positive TEC anomalies and the other hand, a TEC curve that lie below the lower bound show negative TEC anomalies

2.2.4 Statistical Method for the Anomaly Detection

To recognize the anomalies related with Thailand EQ, dual frequency GNSS receiver stations inside and close to the relevant epicentral zones are used in the analysis. we analyze the daily TEC with the thirty-second resolution and DTEC for twenty days prior and ten days next to the main shock day for all GNSS stations. To recognize the TEC anomalies, the confidence bounds are computed by using the below equations. [38]

$$\text{Upper bound (UB)} = m + 2\sigma \quad (\text{x})$$

$$\text{Lower bound (LB)} = m - 2\sigma \quad (\text{xi})$$

where the standard deviation and the median of the TEC time series are denoted by σ and m , respectively.

2.2.5 DTEC measurement:

The dTEC are actually called deviation TEC and explained as the difference between original TEC data and upper bound values. The TEC deviation from the upper bound (DTEC) is computed by subtracting the original TEC from the respective Upper Bound.[39]

$$\text{TEC Deviation (DTEC)} = \text{TEC}_{\text{original}} - \text{Upper Bound} \quad (\text{xii})$$

To differentiate either the ionospheric disturbances related to EQ or geomagnetic storm, this procedure has been used. All calculations were executed by MATLAB. All maps and graphs were also plotted with MATLAB functions.

3 RESULTS

In this thesis, the ionospheric conditions are investigated before the EQ of November 20, 2019 in Thailand (M_w 6.2) happened on UTC = 23:50:43h at the geographical location of Latitude (19.45°N) and Longitude (101.36°E), and shallow hypo central depth 10 km according to USGS as presented in Table 2.1. and shown in Fig. 2.1

3.1 Ionospheric Analysis

To examine the ionospheric anomalies, we have several ionospheric precursors that use to determine either the anomalies are related with seismic events like an earthquake or related with other magnetic phenomena. The main resolution of containing solar and geomagnetic storm conditions in our study is to differentiate whether ionospheric anomalies are induced by the storm or seismic procedure. I have considered four indices related with space weather condition these are:

- Dst
- F10.7
- Kp index
- Ap

The Disturbance storm-time (Dst) index one of parameters relating with the condition of the geomagnetic field. Dst values indicates the intensity of geomagnetic storm activity.

Table 3. 1 Geomagnetic storms classification based on the Dst index

Storm Strength	Great	Severe	Strong	Moderate	Weak	Quiet
Dst values (nT)	< -350	< -200	< -100	< -50	< -30	> -30

The Kp index also called the Planetary index, are used to illustrate the magnitude of geomagnetic storms. Kp index is describing the total disturbance of the Earth's magnetic field in a 3h time interval.

Table 3. 2 Geomagnetic storms classification based on the Kp index

Storm Strength	Extreme	Severe	Strong	Moderate	Minor	Active	Unsettled	Quiet
Kp values	=9	=8	=7	=6	=5	=4	=3	<2

The Ap-index is also called planetary index give a diurnal average level for geomagnetic activity. The Ap index is a measure of the general level of geomagnetic activity over the world of a certain day with units of nT.

Table 3. 3 Geomagnetic storms classification based on the Ap index

Storm Strength	Active	Unsettled	Quiet
AP values	16-29nT	8-15nT	0-7nT

F10.7 is the solar radio flux index with the frequency of 2800 MHz. The wavelength of this index is 10.7 km. This F10.7 is one of the most extensively used indices of solar activity. The F10.7 is an outstanding indicator of solar activity. Normally called the F10.7 index. Its unit is “solar flux units”, (s.f.u.)

Table 3. 4 Solar conditions classification based on the F10.7 index

Storm Strength	extreme	high	moderate	low
F10.7 values	≥ 200 sfu	$(150\text{sfu} \leq \text{F10.7} < 200\text{sfu})$	$100\text{sfu} \leq \text{F10.7} < 150$ sfu	$(\text{F10.7} < 100 \text{ sfu}),$

3.1.1 Kp, Dst, Ap and F10.7 of Thailand Earthquake

The estimation of seismic anomalies related with the Thailand Earthquake, the solar conditions and geomagnetic storm show the EQ occurred during the quiet **geomagnetic activities**. In a geomagnetic quiet period, VTEC also created significant gradually variation. Fig. 3.1 shows the solar conditions and geomagnetic activities for the observational period of the selected EQ, in which a threshold value of 3 is about for the Kp index denoted by a pink line. Furthermore, if a value goes above the threshold line, it is going to be consider as an active geomagnetic activity. And if a Kp index values below the threshold line, it will be considering as a quiet geomagnetic activity. In Fig. 3.1, the solar conditions and geomagnetic storm show the EQ occurred during the quiet **geomagnetic activities** ($K_p < 3$, $\text{Dst} < -10$ nT, $A_p < 7$ and $\text{F10.7} < 100$). However, the Kp is greater than 3 immediately after the EQ day. So the ionospheric anomalies are related with seismic activities (like earthquake).

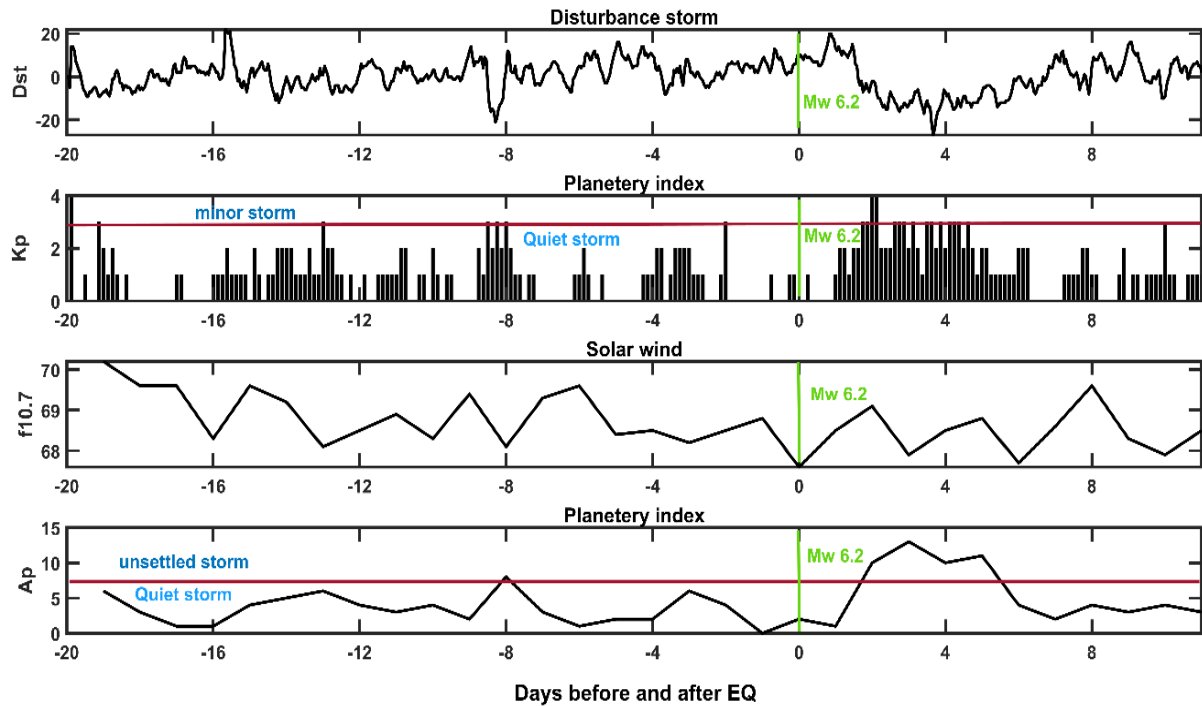


Fig. 3. 1 The Geomagnetic storm indices before and after the (Mw 6.2). Here, the Kp shows active geomagnetic storm after the EQ day.

3.2 Ionospheric TEC Response to Mw 6.2 Thailand EQ

A large EQ of Mw 6.2 occurred at 23:50:43 (UTC) in 32 km ESE of Chaloem Phra Kiat on 20 November 2019. The epicenter was located at Latitude (19.45°N) and Longitude (101.36°E), and shallow hypo central depth 10 km. Fig. 2.2 signifies the geographical positions of the epicenters of EQ at the side of positions of the 3 GNSS stations used for TEC measurements to display the ionospheric conditions prior to the EQ. The outcomes of geomagnetic activity, TEC, and DTEC variations at selected GNSS stations concluded an observational period during October 31-November 30, 2019, as shown in Fig. 3.2-3.4

The estimation of TEC behavior in the ionosphere, the selected GNSS stations are below:

- CMUM (Chiang Mai University, Muang)
- CUUT (Chulalongkorn University, Thailand)
- CPNM (Chumphon University, Muang)

The epicenter is situated at Thailand. GNSS permanent receiver CMUM is positioned at latitude 18.761° and longitude 98.932° in Chiang Mai University, Muang. The CMUM station is very near to epicenter so it shows more anomalies associated with seismic activity and storm, the reason is more energy spreads to close the station. The graph is plotted with TEC data and confidence bounds. In Fig.3.2 the TEC records plotted with confidence bounds which might be UB and LB. In graph the TEC data shows by blue curve lines, upper bound shows by green curve line and the yellow one is for lower bound. The x-axis shows the days that is 20 days earlier and 10 days next to the Earthquake, 0 day show the main shock day. TEC values that lie above the upper bound show positive TEC anomalies and the other hand, a TEC values that lie below the lower bound show negative TEC anomalies

In Fig.3.2 the first window shows about CMUM station. The CMUM station showed the positive TEC anomalies on days -3, -2, -1 before the EQ day and on the EQ day. But, the storm became active after the EQ day; therefore, there are no EQ induced pre-seismic anomalies at the CMUM station. GNSS permanent receiver CUUT is positioned at latitude 13.736° and longitude 100.534° in Chulalongkorn University, Thailand. The CUUT station showed the positive TEC anomalies on days -2, -1 before the EQ day and on the EQ day. The CPNM station also showed the positive TEC anomalies on day -1 before the EQ day

and on the EQ day (Fig. 3.2). Because the storm is active after the EQ day we cannot say confidently about the pre-seismic ionospheric anomalies.

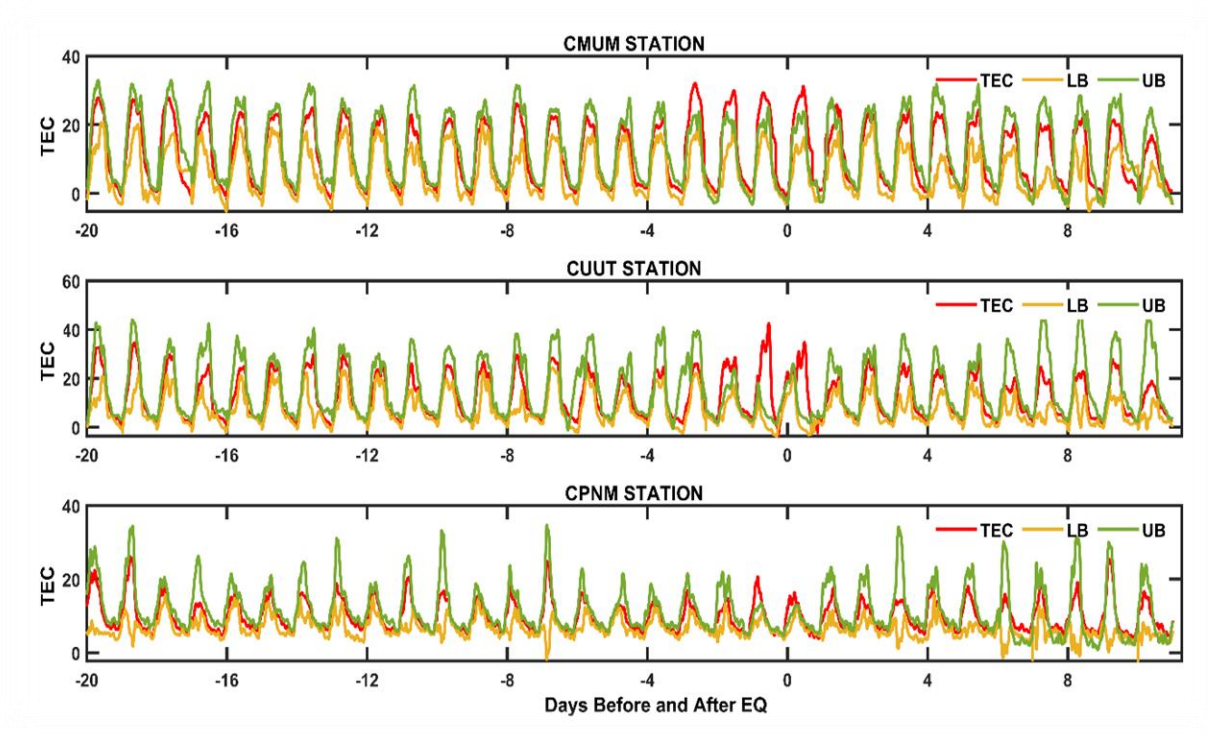


Fig. 3. 2 The TEC measured at three GNSS stations (CMUM, CUUT, CPNM) along with the lower and upper bounds for indicating the seismic ionospheric anomalies.

3.3 Deviation TEC

The TEC deviations (DTEC) from upper bound are calculated for the selected GNSS stations to indicate anomalous ionospheric situations and also filtered out the anomalous days. The DTEC for three IGS stations before and after the EQ can be computed using Eq. (xii) and shown in Figs. 3.3-3.4. We use the temporal and continuous wavelet transform method to find out the DTEC. The positive anomalies are observed from the results of DTEC. But these are storm induced TEC anomalies and no clear evidence of EQ induced ionospheric perturbations came out.

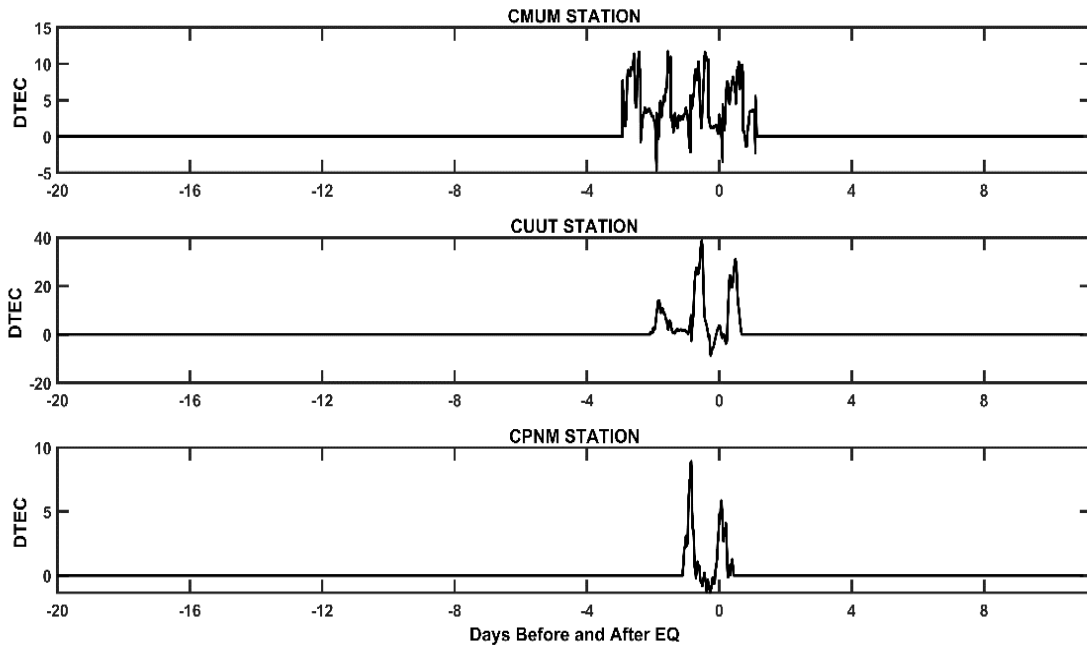


Fig. 3. 3 The DTEC values beyond the upper bounds for all three GNSS stations.

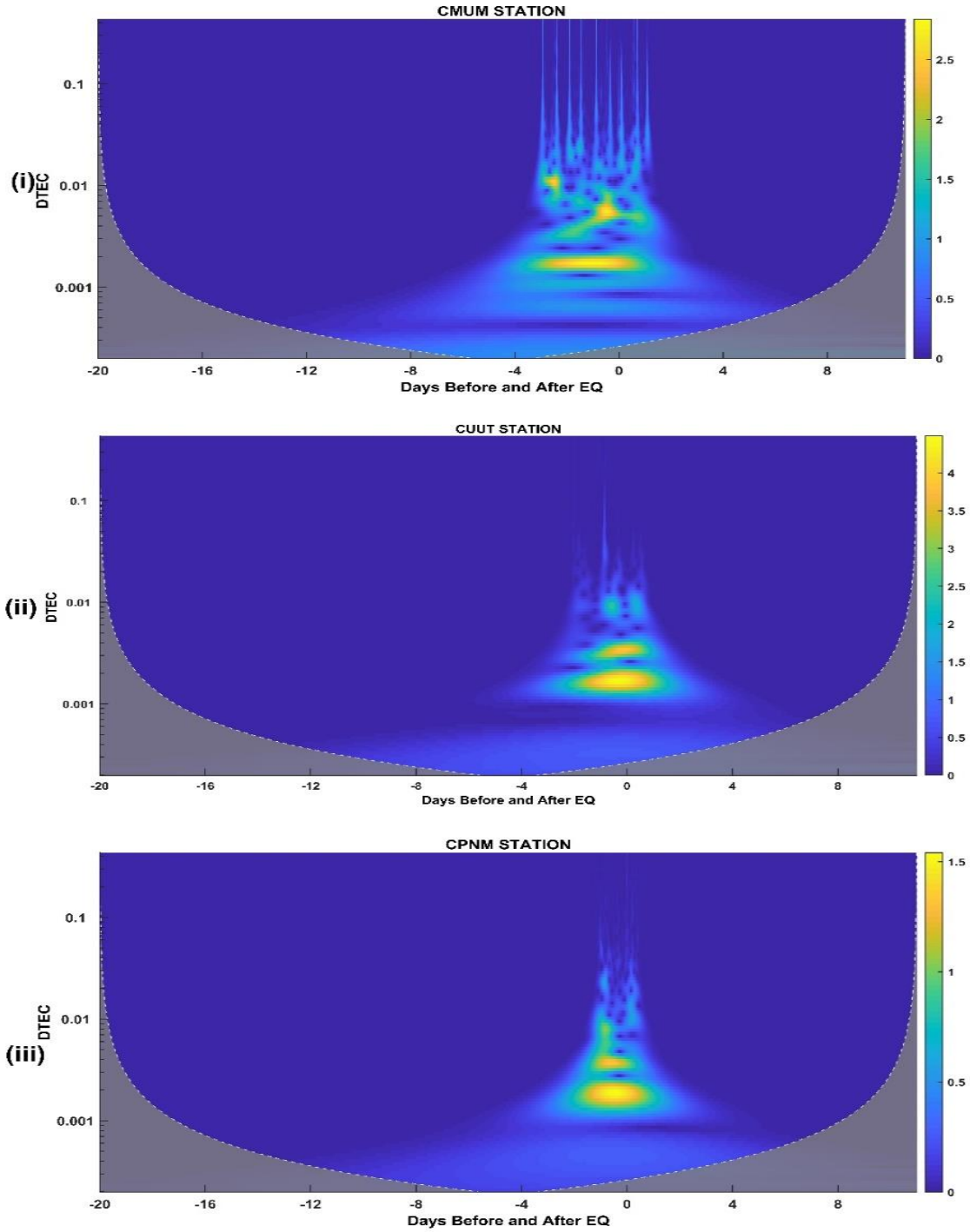


Fig. 3. 4 Wavelet spectrums of DTEC values beyond the upper bounds depicting the TEC disturbances before and on the EQ day for all three Thailand stations (i) CMUM, (ii) CUUT and (iii) CPNM stations.

4 CONCLUSION

We investigated the ionospheric TEC anomalies for the Thailand EQ by analyzing the possible reasons for these anomalies based on solar and geomagnetic storm conditions. The Dst, F10.7, Kp and Ap indices were examined. The studied indices showed the EQ occurred during the quiet geomagnetic activities. Ionospheric TEC data were determined by using GNSS data. The TEC results show that EQ anomalies appeared three days earlier and at the EQ day at some point of quiet geomagnetic storm conditions. But, the storm is active afterward the main shock day and these ionospheric anomalies are related to the storm. As for the computed DTECs, the positive anomalies are greater than the negative anomalies. The data showed the clear ionospheric anomalies over the three days before and on the EQ day.

REFERENCES

- [1] B. Parkinson, K. Gromov, T. Stansell, and R. Beard, “History of Satellite Navigation,” *Proc. Annu. Meet. - Inst. Navig.*, pp. 17–65, 1995.
- [2] J. Gao, “Integration of GPS with remote sensing and GIS: Reality and prospect,” *Photogramm. Eng. Remote Sensing*, vol. 68, no. 5, pp. 447–453, 2002.
- [3] T. H. E. Ieee, “NAVSTAR : Global Positioning System-Ten,” vol. 71, no. 10, pp. 1177–1186, 1983.
- [4] E. Cohen, *Understanding G4*. 2018.
- [5] C. Milner, W. Ochieng, W. Schuster, M. Porretta, and C. Curry, “A regional space segment health monitor for local GPS integrity monitoring,” *J. Navig.*, vol. 64, no. 4, pp. 657–671, 2011, doi: 10.1017/S0373463311000269.
- [6] E. Ali, “Global Positioning System (GPS): Definition , Principles , Errors , Applications & DGPS,” no. April, 2020.
- [7] J. W. Betz, “Binary offset carrier modulations for radionavigation,” *Navig. J. Inst. Navig.*, vol. 48, no. 4, pp. 227–246, 2001, doi: 10.1002/j.2161-4296.2001.tb00247.x.
- [8] S. Revnivkykh, “GLONASS status and modernization,” *25th Int. Tech. Meet. Satell. Div. Inst. Navig. 2012, ION GNSS 2012*, vol. 1, pp. 538–552, 2012.
- [9] N. Limão, “‘국회선진화법’ 에 관한 토론No Title’,” *입법학연구*, vol. 제13집 1호, no. May, pp. 31–48, 2016.
- [10] F. Breu, S. Guggenbichler, and J. Wollmann, “GLONASS Overview,” *Vasa*, pp. 1–

- 7, 2008, [Online]. Available:
<http://medcontent.metapress.com/index/A65RM03P4874243N.pdf>.
- [11] E. S. Lohan, “GALILEO Positioning Technology,” *GALILEO Position. Technol.*, vol. 182, pp. 1–3, 2015, doi: 10.1007/978-94-007-1830-2.
 - [12] “China ’ s BeiDou Navigation Satellite System,” no. June, 2016.
 - [13] CSNO, “BeiDou Navigation Satellite System Signal In Space Interface Control Document Search and Rescue Service (Version 1.0),” 2020.
 - [14] X. Wang, “gLAB upgrade with BeiDou Navigation System Signals,” no. July, p. 104, 2015.
 - [15] M. Bakula, “An approach to reliable rapid static GNSS surveying,” *Surv. Rev.*, vol. 44, no. 327, pp. 265–271, 2012, doi: 10.1179/1752270611Y.0000000038.
 - [16] D. Borio, F. Dovis, H. Kuusniemi, and L. Lo Presti, “Impact and Detection of GNSS Jammers on Consumer Grade Satellite Navigation Receivers,” *Proc. IEEE*, vol. 104, no. 6, pp. 1233–1245, 2016, doi: 10.1109/JPROC.2016.2543266.
 - [17] A. Rustamov, N. Gogoi, A. Minetto, and F. Dovis, “Assessment of the Vulnerability to Spoofing Attacks of GNSS Receivers Integrated in Consumer Devices,” *2020 Int. Conf. Localization GNSS, ICL-GNSS 2020 - Proc.*, pp. 1–6, 2020, doi: 10.1109/ICL-GNSS49876.2020.9115489.
 - [18] M. Jonáš, “GNSS Integrity for Railway Transportation,” *Trans. Transp. Sci.*, vol. 4, no. 4, pp. 183–192, 2011, doi: 10.2478/v10158-011-0016-6.
 - [19] C. Facchinetti, L. Ansalone, and A. Tuoizzi, “Trends in GNSS Italian application

- scenarios in transportation,” *2015 Int. Assoc. Institutes Navig. World Congr. IAIN 2015 - Proc.*, no. October, pp. 20–23, 2015, doi: 10.1109/IAIN.2015.7352256.
- [20] C. Rizos, “Locata : A Positioning System for Indoor and Outdoor Applications Where GNSS Does Not Work,” no. March, pp. 12–14, 2013.
- [21] Y.-W. Lee, Y. Suh, and R. Shibasaki, “Ajax GIS application for GNSS availability simulation,” *KSCE J. Civ. Eng.*, vol. 11, no. 6, pp. 303–310, 2007, doi: 10.1007/bf02885901.
- [22] J. Bremer, “Trends in the ionospheric E and F regions over Europe,” *Ann. Geophys.*, vol. 16, no. 8, pp. 986–996, 1998, doi: 10.1007/s00585-998-0986-9.
- [23] W. I. Axford, “The formation and vertical movement of dense ionized layers in the ionosphere due to neutral wind shears,” *J. Geophys. Res.*, vol. 68, no. 3, pp. 769–779, 1963, doi: 10.1029/jz068i003p00769.
- [24] D. F. Webb and T. A. Howard, “Coronal Mass Ejections : Observations Imprint / Terms of Use,” *Living Rev. Sol. Phys.*, vol. 9, no. 3, 2012.
- [25] S. M. Radicella and B. Nava, *Empirical ionospheric models*. Elsevier Inc., 2019.
- [26] G. Seeber, “Satellite Geodesy, Walter de Gruyter,” *Berlin Shen Y, Chen Q, Hsu H, Zhang X, Lou L A Modif. short arc approach Recover. gravity eld Model*, 2003.
- [27] J. A. Klobuchar, “Ionospheric Time-Delay Algorithm for Single-Frequency GPS Users,” *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-23, no. 3, pp. 325–331, 1987, doi: 10.1109/TAES.1987.310829.
- [28] R. Citation, “Analysis of Total Electron Content (TEC) Variations in the Low- and

Middle-Latitude Ionosphere,” 2009.

- [29] M. M. Hoque, N. Jakowski, and J. A. Cahuasqui, “Fast ionospheric correction algorithm for galileo single frequency users,” *2020 Eur. Navig. Conf. ENC 2020*, pp. 1–10, 2020, doi: 10.23919/ENC48637.2020.9317502.
- [30] S. M. Radiceila, “The NeQuick model genesis, uses and evolution,” *Ann. Geophys.*, vol. 52, no. 3–4, pp. 417–422, 2009.
- [31] R. Píriz, P. Roldán, R. Golcz, C. Moriana, and J. Leute, “Performance of the NeQuick G iono model for single-frequency GNSS timing applications,” *2016 Eur. Freq. Time Forum, EFTF 2016*, pp. 2–5, 2016, doi: 10.1109/EFTF.2016.7477769.
- [32] B. Zhao, M. I. N. Wang, T. A. O. Yu, and G. Xu, “Ionospheric total electron content variations prior to the 2008 Wenchuan Earthquake,” vol. 31, no. 13, pp. 3545–3557, 2010, doi: 10.1080/01431161003727622.
- [33] N. Bhattarai, N. P. Chapagain, and B. Adhikari, “Total Electron Content and Electron Density Profile Observations during Geomagnetic Storms using COSMIC Satellite Data,” *Discovery*, vol. 52, no. 250, pp. 1979–1990, 2016, [Online]. Available: <http://omniweb.gsfc.nasa.gov/ow.html>.
- [34] A. Kiyani *et al.*, “Seismo ionospheric anomalies possibly associated with the 2018 Mw 8.2 Fiji earthquake detected with GNSS TEC,” *J. Geodyn.*, vol. 140, no. January, p. 101782, 2020, doi: 10.1016/j.jog.2020.101782.
- [35] F. Zhu, Y. Wu, Y. Zhou, and Y. Gao, “Temporal and spatial distribution of GPS-TEC anomalies prior to the strong earthquakes,” *Astrophys. Space Sci.*, vol. 345, no.

- 2, pp. 239–246, 2013, doi: 10.1007/s10509-013-1411-8.
- [36] Y. Yuan, C. C. Tscherning, P. Knudsen, G. Xu, and J. Ou, “The ionospheric eclipse factor method (IEFM) and its application to determining the ionospheric delay for GPS,” *J. Geod.*, vol. 82, no. 1, pp. 1–8, 2008, doi: 10.1007/s00190-007-0152-2.
- [37] I. P. Dobrovolsky, S. I. Zubkov, and V. I. Miachkin, “Estimation of the size of earthquake preparation zones,” *Pure Appl. Geophys. PAGEOPH*, vol. 117, no. 5, pp. 1025–1044, 1979, doi: 10.1007/BF00876083.
- [38] M. Shah, M. A. Tariq, J. Ahmad, N. Abbas, and S. Jin, “Seismo ionospheric anomalies before the 2007 M7 . 7 Chile earthquake from GPS TEC and DEMETER,” *J. Geodyn.*, vol. 127, no. August 2018, pp. 42–51, 2019, doi: 10.1016/j.jog.2019.05.004.
- [39] J. Y. Liu *et al.*, “Seismoionospheric GPS total electron content anomalies observed. before the 12 May 2008 Mw7.9 Wenchuan earthquake,” *J. Geophys. Res. Sp. Phys.*, vol. 114, no. 4, pp. 1–10, 2009, doi: 10.1029/2008JA013698.